Monnit MODBUS TCP Utility

Monnit MODBUS TCP Utility - User Guide

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1. Getting Started

Application Overview & Purpose

The Monnit ALTA MODBUS TCP Utility is a high-performance local software solution engineered specifically for the Monnit ALTA Ethernet Gateway 4. Its primary mission is to bypass cloud dependencies by communicating directly with the gateway's MODBUS TCP Interface on Port 502. This allows for near real-time data acquisition in environments where internet connectivity is restricted, unreliable, or entirely absent.

By acting as a **Protocol Bridge**, the utility transforms raw, binary register data into human-readable engineering units. This localized approach ensures that your sensitive environmental data never leaves your facility, providing a critical layer of security for industrial, medical, and high-security installations.

Security and Reliability

In many industrial sectors—such as pharmaceutical storage or data center management—relying on a third-party cloud provider introduces a point of failure and a potential security vulnerability. This utility eliminates those risks by maintaining a 100% local data loop. If your internet goes down, your monitoring, alerting, and data logging continue uninterrupted.

Important Limitation: Sensor Configuration

It is vital to understand that this utility operates as a **Data Consumer**. The MODBUS TCP interface provided by the gateway hardware is strictly "Read-Only." Consequently, hardware-level changes must still be managed via Monnit's iMonnit software (either the cloud or Enterprise version).

Tasks that MUST be performed in iMonnit:

- **Commissioning:** Adding new sensors or removing decommissioned ones from the gateway.
- **Network Timing:** Adjusting the Heartbeat Interval (how frequently a sensor transmits).
- **Hardware Alerts:** Setting "Aware State" thresholds that trigger the sensor's internal alarm logic.
- **Calibration:** Applying "Linear" or "User" calibration coefficients to the sensor hardware itself.

2. Setup Wizard

The Setup Wizard is designed to handle the "Heavy Lifting" required to prepare a factory-default gateway for local MODBUS communication.

Step-by-Step Discovery

- **Multi-Protocol Scanning:** The utility doesn't just check one IP at a time; it uses a robust four-pass scanning strategy:
    1. **mDNS (Avahi/Bonjour):** Listens for multicast announcements.
    2. **DNS Reverse Lookups:** Scans hostnames for "sensgateway" matches.
    3. **HTTP Probing:** Checks Port 80 for the Monnit configuration interface.
    4. **MODBUS Probing:** Checks Port 502 for active responders.
- **Gateway Verification:** When a device is discovered, the utility performs a "Handshake." It parses the device's internal status pages to verify it is an authentic ALTA Ethernet Gateway 4 and checks the firmware version (v1.1.0.6+ is required).

Critical Configuration Checks

1. **Unlock Status:** Monnit gateways are often sold in a "Locked" state to ensure they report to the cloud. To work locally, the gateway must have a "Modbus Unlock" applied. The wizard will detect this and provide guidance on how to apply an unlock code.
2. **Default Server (iMonnit) Conflict:** If a gateway is on a private network and cannot reach the internet, it will spend excessive resources trying to find "iMonnit." This can cause the internal sensor network to go inactive. The wizard recommends disabling the "Default Server" interface to ensure 24/7 local stability.

3. Dashboard Overview

The Dashboard is a centralized hub for all your environmental data, organized into logically segmented tabs:

- **Monnit Devices:** This tab features "Live Cards." Each card updates dynamically as the gateway receives new data from the sensors.
    - **Green Indicator:** Standard operation; data is current and within limits.
    - **Red Indicator:** The sensor is in an "Aware State" (e.g., Temperature exceeded a limit).
    - **Yellow Indicator:** Maintenance required (Low battery or failing signal strength).
- **External Devices:** A powerful feature that displays data ingested from non-Monnit sources via plugins. This allows you to view MQTT, Webhook, or BACnet data side-by-side with your Monnit sensors, creating a unified monitoring pane.
- **Gateway Settings:** Shows high-level health metrics of the hardware, including firmware version, MAC address, and current IP lease.
- **Portal Settings:** The "Brain" of the application where you configure how the *software* behaves, rather than the hardware.

4. Device Details & History

Deep-diving into your data is essential for identifying trends and long-term environmental patterns.

- **History Tab:** This tab maintains a high-fidelity record of every check-in. The table includes timestamps, raw data values, and even the RSSI (Signal Strength) at the time of transmission.

  - *Pro Tip:* Use the Date Pickers to isolate "Events"—such as a power outage or a cooling system failure—to see exactly how the environment responded.

- **Charts Tab:** Visualizes data using interactive line charts. You can toggle specific data points (like Battery Voltage vs. Temperature) to see if battery performance is being impacted by extreme cold.
- **Alert History:** This is a localized audit log. It tracks every time the *utility software* triggered a notification. This is separate from the gateway's internal logs and is invaluable for compliance reporting and incident post-mortems.

5. Gateway Management

The utility is capable of storing a library of many gateways, but it focuses its resources on **one active polling connection** at a time to ensure maximum data integrity and prevent network congestion.

- **Switching Gateways:** To move between different zones (e.g., "Warehouse A" to "Warehouse B"), navigate to Gateway Management, select the new gateway, and activate it. The polling engine will automatically re-initialize for the new register map.
- **The Reboot Cycle:** Most settings changes (like changing a Subnet Mask or a Server IP) require the gateway hardware to write to its flash memory and reboot. This utility manages that process, providing a 30-60 second countdown and automatically re-establishing the MODBUS connection once the device is back online.

6. Alert Configuration

While the sensors have internal alarms, the utility provides a powerful **Software-Side Alerting** layer that runs on your server/PC.

Notification Methods

- **SMTP (Email):** Supports standard encryption (SSL/TLS). You can use public providers like Gmail (Port 587/465) or internal corporate relay servers.
- **Textbelt (SMS):** A simplified SMS delivery service. This is ideal for critical "off-hours" notifications where an email might be missed.

Logic & Suppression

- **Thresholds:** You can set "Above" or "Below" triggers.

- **Repeat Intervals:** To avoid "Notification Fatigue," you can set a cooldown. If a sensor stays in an alert state, the utility won't "spam" your inbox; it will only re-notify you after your specified interval (e.g., every 60 minutes) has passed.

7. Application Settings & Automation

This section houses the advanced "Set and Forget" features that make the utility a robust industrial tool.

Auto Backup (Redundancy & Security)

The application's database contains your historical data. To prevent loss:

- **Automatic Snapshots:** Configure the frequency (Hourly to Monthly).
- **Retention:** Keep up to 10 historical versions of your database.
- **Storage Location:** Backups are stored in the local backups directory. In the event of a PC failure, you can simply move the sensor_data.db file to a new installation to restore all your history and settings.

Auto Downloader (Compliance Reporting)

For industries like pharmaceuticals (CDC/VFC) or food storage (HACCP), regular data logs are a legal requirement.

- **PDF Exports:** Creates a non-editable, professional report that is ready for auditors.
- **CSV Exports:** Ideal for data analysts who want to import Monnit data into Excel or PowerBI.
- **Scheduled Delivery:** Files are silently generated at your chosen time (e.g., 12:00 AM every Sunday) and saved to Documents/Monnit_Data_Export.

8. System Tray & Background Mode

To ensure 24/7 monitoring, the utility is built with a "Background-First" philosophy.

- **The "X" Behavior:** When you click the Close (X) button, the window disappears, but the **Node.js polling engine remains active**. You will continue to receive Email/SMS alerts even if the UI is not visible.
- **Tray Controls:** Right-click the icon in your system tray to quickly see the status of the polling engine or to fully quit the application when maintenance is required on the host PC.

9. Advanced: MODBUS TCP Interface

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For system integrators and PLC programmers, this section provides the specific technical details required for external integration with the gateway hardware.

Network Configuration

The ALTA Ethernet Gateway 4 acts as a **Modbus Server**.

- **Default Port:** 502
- **Unit ID:** 1 (Static)
- **Addressing:** 0-Based (Standard Modbus Protocol). Note that some PLCs use 1-based addressing; in such cases, add 1 to the register addresses listed below.

Gateway Status Registers (Holding Registers)

These registers provide high-level status of the gateway itself.

Register	Description	Data Type
0-1	Gateway ID	32-bit Unsigned (High/Low Word)
2	Firmware Version	16-bit (e.g., 106 = 1.0.6)
3	Radio Status	16-bit (Bitmask for radio health)
4	Sensor Count	16-bit (Number of sensors in network)

Sensor Data Registers (100+)

The gateway allocates a **16-register block** for every sensor slot.

- **Slot 1:** 100 - 115
- **Slot 2:** 116 - 131
- **Formula:** Base Address = 100 + (Slot - 1) * 16

The 16-Register Map

Offset	Description	Interpretation
+0-1	Sensor ID	32-bit Unsigned
+2	Sensor Type	See "Common Sensor Types" below
+3	Data Age	Seconds since the last heartbeat

+4	Active State	0 = Inactive, 1 = Active
+5	Aware State	0 = Normal, 1 = Triggered Alarm
+6	Battery Voltage	Centivolts (e.g., 320 = 3.20V)
+7	RSSI	Signal strength percentage (0-100)
+8-15	Data Payload	8 Words of sensor-specific readings

Common Sensor Types & Parsing

Data in registers +8 through +15 must be parsed based on the **Sensor Type** found in offset +2.

- **Type 2: Temperature**
  - Reg+8: 16-bit Signed Integer. Value is 10x Celsius.
  - *Example:* 235 = 23.5°C. For negative values, apply 2's complement (if value > 32767, then val = val - 65536).
- **Type 16: Humidity**
  - Reg+9: 16-bit Unsigned Integer. Represents % Relative Humidity.
- **Type 43: AC Current (3-Phase/1-Phase)**
  - Reg+8 (Low) and Reg+9 (High): Combine into a 32-bit integer.
  - *Formula:* Amps = (Low + (High << 16)) / 1000.
- **Type 4: Water Detect / Type 3: Dry Contact**
  - Reg+8: 0 = Dry/Open, 1 = Wet/Closed.

10. Application Updates

Maintenance is handled via an integrated update engine.

- **Background Checks:** The utility checks once per week for security patches.
- **Safety First:** Updates are downloaded in a "Staging" area. The application will never force a restart while you are actively configuring a gateway; it waits for you to click "Restart & Install."

11. Accessing Help Documentation

The utility includes a full offline documentation suite for secure environments. In the **Portal Settings** tab, use the **Click for Help Documentation** link to access:

- **TROUBLESHOOTING.pdf:** Solutions for MODBUS timeouts and IP conflicts.
- **SECURITY_ARCHITECTURE.pdf:** Details on data encryption and local database security.
- **CHANGELOG.pdf:** A history of every feature and protocol update.